FAKE NEWS DETECTION USING NATURAL LANGUAGE PROCESSING (NLP) :

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PROBLEM DESCRIPTION :

The problem is to develop a fake news detection model using a Kaggle dataset. The goal is to distinguish between genuine and fake news articles based on their titles and text. This project involves using **Natural Language Processing (NLP)** techniques to preprocess the text data, building a **machine learning model** for classification, and evaluating the model's performance.

Now-a-days with the ease of content creation and due to the wide spread access of internet all over the world , fake news has become a major problem . This type of fake news may be misleading and may give the wrong information about a event which is dangerous.

The spread of fake news erodes trust in media and information sources, making it increasingly difficult for individuals to make informed decisions based on accurate information.

Fake news can influence public opinion, elections, and social dynamics, leading to harmful consequences for individuals and society as a whole .

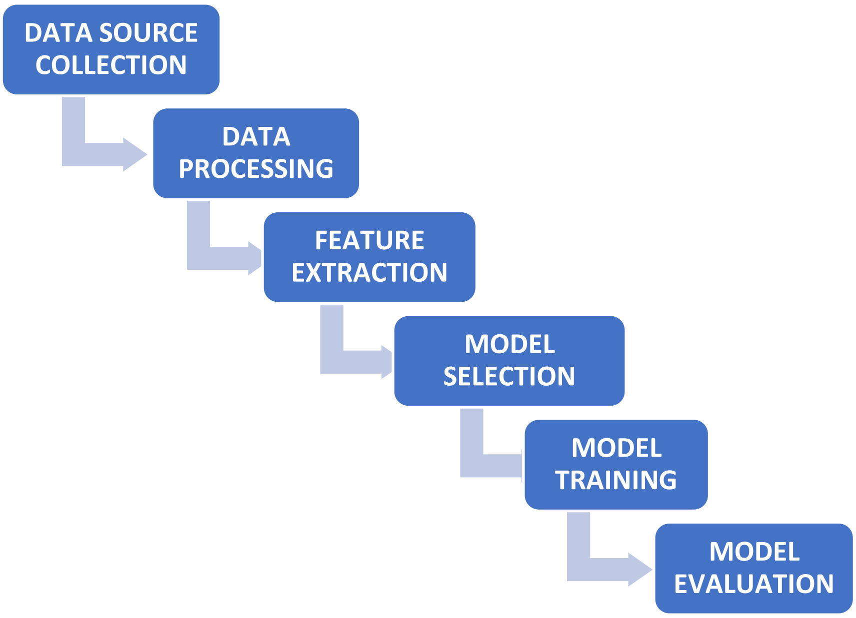
So this project will be focused on detecting the fake news in the internet with the help of the NLP and by building a machine learning model using **Python .**

DESIGN THINKING :

The design thinking for this project includes **six steps** , they are

1. Data source collection
2. Data Processing
3. Feature extraction
4. Model Selection
5. Model Training
6. Model Evaluation

Let us see about the each phase in detail.



1. DATA SOURCE COLLECTION :

In this step **the required dataset** for the fake news detection is collected .For this process the classical dataset for fake detection from the Kaggle is taken . This dataset consists of the true news and the fake news and so by using this dataset it will be effective for the training of the ML model.

The link for the dataset we will be using for this project is : **https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset**

2. DATA PROCESSING :

In this step the **dataset is cleaned** and the null values are removed with the help of the **python** language .The **pandas** module used for loading the dataset and for cleaning the process . Thus in this process the clean data for the training is prepared.

In the cleaning process **the null values** will be removed or it will be filled with repeated values and the duplicate data will be deleted.

Tools used : **Pandas** , **Numpy** and **scikit-learn** modules in python .

3. FEATURE EXTRACTION :

In this step the words are converted into the numerical features. This can be done by utilizing the techniques like **TF-IDF** (Term Frequency-Inverse Document Frequency) or **word embeddings** to convert text into numerical features.

4. MODEL SELECTION :

The **suitable classification algorithm** for training the model is chosen in this step. There are several classificatioin algorithms like **Logistic regression ,Random forest , neural networks ,SVM** ,etc .

We will use the **Artificial neural networks** for this project.

Linear regression will be used to establish a linear relationship between the input and the target variable .

The **Support Vector machine (SVM**) will be used to perform regression tasks , capable of handling non-linear relationships .

5. MODEL TRAINING:

In this step the **preprocessed and cleaned data** is used to train the model with the help of the classification algorithms . The training process involves **finding the underlying patterns** in the dataset and enabling the model to perform **accurate predictions** ,etc .

The model should be trained with the **various forms of the preprocessed data** to improve its accuracy . The more data we feed to the model the **more accurate results** will be generated by the model .

6. MODEL EVALUATION :

In this step the trained model is evaluated with **various inputs** . The model is evaluated based on the metrics like **accuracy, precision, recall, F1-score, and ROC-AUC.**

If any error occurs during the evaluation of the model then the model is again moved to the Model Training phase .

RESULTS :

The results of the model will be evaluated by given the **various input datas based on different criteria .** If the accuracy rate is less then the input and output are **again used as the training data** to the model .

FURTHER PROJECT IMPLEMENTATIONS:

Using the above six steps in the design thinking the project will be developed with the **Agile methodology** and the errors and issues in each phase will be cleared at the respective phases itself .

With help of the help of the Version Control Systems like **Github** the project will be tracked and the final project will be developed successfully .

TOOLS / MODULES USED:

The tools and some of the modules uses in this project are

1. **Python** language
2. **Numpy , pandas , scikit-learn , matplotlib** modules
3. Classification algorithms like **linear regression , random forest and SVM**.